MSE 312 — Protecting the MCU on a DC-motor drive

Why the Arduino output stage died, and the bulk cap + protective measures that keep the replacement alive. For the TB6568KQ / SN7400 motor drive.

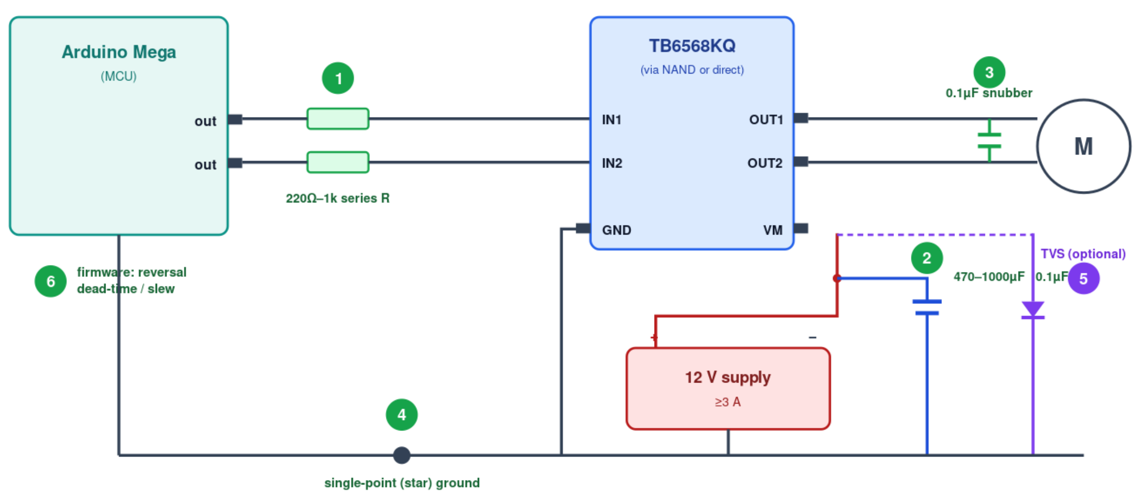
What killed the board

The symptom was: 5 V rail alive, code running, analog and digital **inputs** fine, but every digital **output** stuck near 0.3 V even on a bare pin — the GPIO output drivers were destroyed while the rest of the chip survived. The cause is transient over-stress. On a hard CW↔CCW reversal with no bulk capacitor, the motor dumps back-EMF, the 12 V rail rings and sags, and the shared ground bounces. Those transients reach the logic nodes tied to the MCU pins; when a pin is forced above Vcc or below GND, its internal clamp diodes conduct, and with no bulk cap to absorb the spike the current through them cooks the output driver. Inputs are high-impedance and better protected, so they live on — hence a board that can still 'receive' but not 'drive'.

Protection map

MCU protection map — where each measure goes

Numbered items match the table in the document · green = added protection



The six measures

#	Measure	Value / part	Where it goes	What it stops
1	Series resistors	220 Ω – 1 kΩ, 1/4 W (one per line)	in series, each MCU pin → driver/NAND input	limits current into the pin's clamp diodes — the single best pin saver
2	Bulk cap + ceramic	470–1000 µF electrolytic (≥25 V) ■ 0.1 µF ceramic	across VM ↔ GND right at the driver pins	absorbs inrush/reversal sag and fast spikes on the 12 V rail
3	Motor snubber	0.1 µF ceramic (≥50 V)	across the motor terminals (OUT1 ↔ OUT2)	kills brush + switching EMI at the source
4	Star ground	short, thick wire	all grounds meet at ONE point (supply –)	stops motor return current from bouncing the MCU's ground reference
5	TVS diode (optional)	bidirectional, ~15–18 V standoff (e.g. 1.5KE18CA / SMBJ15CA)	across VM ↔ GND	clamps large transient spikes on the motor rail
6	Firmware dead-time	a few ms coast/brake + duty slew	at every CW↔CCW change, in code	cuts the di/dt that creates the transient in the first place

Bulk cap — why it matters most

The bulk capacitor is the heart of the fix. It's a local energy reservoir sitting right across the motor supply. When the motor demands a sudden current surge (startup, stall, or a direction reversal), the cap supplies that charge instantly instead of letting the supply voltage collapse and ring. That keeps VM stable, keeps the ground quiet, and removes the transient that was reaching the MCU. Rules: use a **470–1000 µF electrolytic rated ≥25 V**, place it **physically at the driver's VM and GND pins** (not back at the supply), observe polarity, and always pair it with a **0.1 µF ceramic** right beside it — the electrolytic is too slow for the fast edges, the ceramic handles those.

Shopping list (per drive)

Qty	Part	Spec	Notes
1	Electrolytic capacitor	1000 µF, 25 V	bulk energy store on 12 V rail; mind polarity
2	Ceramic capacitor	0.1 µF, 50 V, X7R	one at VM/GND, one across motor terminals
3	Resistor	470 Ω, 1/4 W	series in each MCU→driver-input line
1	TVS diode (optional)	1.5KE18CA (bidir)	clamp on VM; optional but cheap insurance
1	Bench/regulated supply	12 V, ≥3 A	headroom for inrush; better than a small wall adapter

Firmware measures (measure 6)

Hardware alone isn't enough — don't command instantaneous full-power reversals. At every direction change, drop to zero drive, hold a brief **dead-time** (a few ms of coast or brake) so the winding current decays, then ramp up the new direction with a **duty slew** (e.g. limit to ~10 V/s) rather than a step. This keeps di/dt and the resulting spike small. Your existing soft-start covers the ramp; add the dead-time at the CW↔CCW seam.

Commissioning checklist

- Wire the star ground first; confirm Arduino GND, driver GND, supply– and cap– all meet at ONE node.
- Install the bulk electrolytic (correct polarity!) + 0.1 µF ceramic at the driver VM/GND pins.
- Add the 0.1 µF motor snubber and the series resistors in the MCU→input lines.
- Power the motor rail from a current-limited supply first; watch VM on a scope during a slow CW↔CCW sweep — it must not dip toward 8 V (UVLO) or ring hard.
- Only after VM is clean, enable full-speed operation and the firmware reversal dead-time.
- If a pin ever reads ~0.3 V on HIGH again, stop — a transient is still getting through; recheck grounds, cap placement, and series resistors.